

Confidence-Based Self-Training for EMG-to-Speech: Leveraging Synthetic EMG for Robust Modeling

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Abstract—Voiced Electromyography(EMG)-to-Speech (V-ETS) models reconstruct speech from muscle activity signals, facilitating applications such as neurolaryngologic diagnostics. Despite its potential, the advancement of V-ETS is hindered by a scarcity of paired EMG-speech data. To address this, we propose a novel Confidence-based Multi-Speaker Self-training (CoM2S) approach, along with a newly curated Libri-EMG dataset. This approach leverages synthetic EMG data generated by a pre-trained model, followed by a proposed filtering mechanism based on phoneme-level confidence to enhance the V-ETS model through the proposed self-training techniques. Experiments demonstrate our method improves phoneme accuracy, reduces phonological confusion, and lowers word error rate, confirming the effectiveness of our CoM2S approach for V-ETS. In support of future research, we will release the codes and the proposed Libri-EMG dataset—an open-access, time-aligned, multi-speaker voiced EMG and speech recordings.

Index Terms—Voiced EMG-to-speech, confidence-based self-training, multi-speaker EMG-speech dataset

I. INTRODUCTION

Voiced Electromyography-to-Speech (V-ETS) aims to reconstruct speech from muscle activity signals, facilitating the interpretability and controllability by simultaneously capturing articulatory muscle signals and speech [1]. V-ETS greatly supports AI in healthcare applications such as neurolaryngologic diagnostics [2], research endeavors in manipulable speech generation [3], [4] and articulatory-to-acoustic decoding [5], [6]. Unlike silent ETS, which maps unvoiced muscle activity to silent speech and lacks direct exposure to audible speech, V-ETS captures a stronger relationship between muscle signals and actual speech. With audible speech informed, V-ETS enables more precise modeling of speech-related neuromuscular activity. While silent ETS may benefit from speech-aligned cues informed by V-ETS, the latter offers complementary advantages for studying speech production and phoneme articulation. Therefore, V-ETS is essential for advancing our understanding of the physiological basis of speech production and phoneme articulation.

Although research on Electromyography-to-Speech (ETS) has been growing [1], [7]–[23], the availability of EMG-speech datasets remains limited due to labor-intensive and costly data

collection from human participants. Currently, open-access voicing EMG-speech datasets include only 2 hours of recordings from [22], 20 hours from [7], and 9.5 hours from [17]. More critically, differences in signal recording configurations across datasets make them incompatible with one another. For instance, [17] used a high-density setup with 40 electrodes, while [7] employed only 8, and [22] recorded with just 6. These inconsistencies introduce signal mismatches. Such data scarcity and incompatibility poses significant challenges, particularly in machine learning and deep learning modeling which may require consistent and meaningful EMG inputs.

To address this, some studies have explored data augmentation techniques to improve model performance. For instance, [12] investigated self-learning and active-learning strategies to expand datasets, demonstrating that human-in-the-loop corrections significantly enhanced model performance. [13] introduced SU-ETS, a model that predicts speech units (SUs) [24] from EMG signals for speaker-independent synthesis by incorporating a voice conversion model. However, these augmentation approaches primarily rely on reusing existing speech content without introducing new phonological knowledge (e.g., transforming the same sentence into different voices without changing its wording [13] or filtering the original dataset before retraining [12])—and often still require additional human effort, such as re-recording speech to improve data quality [12]. Moreover, existing ETS models have yet to surpass the best published voiced WER of 23.3% [8], highlighting the need for alternative methods to increase dataset size and new training strategies.

A promising approach to address data limitation is self-training, a semi-supervised learning technique that has proven effective in other fields [25]–[27], where self-training fundamentally involves using the model’s own predictions (or outputs) to creating additional training data (pseudo-labels). However, despite its success in other tasks, self-training remains largely unexplored in the ETS domain, leaving a significant gap in research that this work seeks to address.

To bridge this gap, we propose a novel data augmentation approach and training strategy that combines self-training to extract high-quality EMG-speech time-aligned data from a large repository of open-access speech resources, LibriSpeech

[28]. Our method, **Confidence-based Multi-Speaker Self-training (CoM2S)** for V-ETS, leverages a pre-trained generator model [21] to generate EMG features aligned with multi-speaker speech and employs a confidence-based self-training strategy to filter high-quality synthetic samples. This approach effectively expands the available data without requiring additional costly EMG recordings while mitigating data mismatch issues by including session embeddings.

The main contributions of this work are as follows:

- **Threshold-Tuned Self-Training:** We incorporate self-training into the V-ETS domain by systematically evaluating phoneme accuracy thresholds to optimize the quality-quantity tradeoff of synthetic EMG data;
- **Open-Source Dataset:** We introduce Libri-EMG, an 8.3-hour open-access high quality multi-speaker voicing dataset, expanding multi-speaker EMG data to support further research in EMG-based speech modeling [29];
- **Extensive Experiments:** We analyze the impact of different training strategies and data ratios, demonstrating that a 1:1 real-to-synthetic mixing achieves optimal performance and outperforms the best published [8] voiced WER of 23.3%.

II. METHODS

Given the scarcity of EMG-speech time-aligned data, in this work, we investigate whether synthetic data—paired with confidence-based filtering—can be used effectively in a self-training approach to V-ETS modeling.

Fig. 1 illustrates the CoM2S approach pipeline (top left), baseline architecture (top right), and CoM2S V-ETS inference pipeline (bottom). Our CoM2S approach begins by generating time-aligned EMG features from speech, simulating the EMG modality across diverse speakers. We then employ a self-training approach in which the baseline model together with a confidence-based filter generates pseudo-labels for synthetic inputs. We further explore how filter thresholds affect model performance, and whether small but high-confidence subsets may outperform larger, less filtered ones. Additionally, we investigate the impact of mixing filtered synthetic data with real EMG data in various proportions during self-training. Finally, we propose a train-from-scratch approach using real paired EMG-speech inputs and synthetic ones to evaluate synthetic multi-speaker EMG-speech data quality that jointly processes real and synthetic paired EMG-speech data, enabling direct comparison of their contributions to model learning.

We aim to identify optimal strategies for leveraging synthetic data and confidence filtering to enhance V-ETS model in low-resource scenarios.

A. Generation of Time-aligned EMG Features from Multi-Speaker Speech

We adopted the pretrained generator from a generator model [21], which builds upon [30]. The EMG generator is conditioned on speech content representations extracted by voice conversion (VC) models, allowing it to take speaker-independent Soft Speech Units (Soft SUs) as input [24] [30].

While the generator does not differentiate between speakers at the voice level, it incorporates learnable session embeddings to compensate for variations in electrode configurations across different EMG recording sessions. Since our CoM2S approach is not constrained to a specific recording parameter, we assign session indices randomly but evenly to match the distribution used in our baseline ETS model. By conditioning the generator on speaker-independent Soft SUs [13], we maintain speaker invariance at the speech content level while leveraging learnable session embeddings to address variability in electrode configurations, aligning with that of our baseline ETS model.

B. Self-training Pseudo-Labeling with Confidence-Based Filtering

We propose to utilize the pretrained generator [21] to generate speaker-independent synthetic EMG features conditioned on multi-speaker speech. However, these generated EMG features are not guaranteed to be speaker-consistent or fully faithful to natural EMG patterns. As such, although the accompanying speech is real and labeled, the pair (synthetic EMG-real speech) is not ground-truth-aligned in the conventional sense. Therefore, we treat the real speech (or its derived features such as MFCCs or phoneme labels) as pseudo labels for the synthetic EMG input. By applying confidence-based filtering, we select only the synthetic EMG-speech pairs for which the transduction model produces confident outputs. This filtering acts as a form of pseudo-label validation, ensuring that only plausible synthetic EMG inputs with reliable label alignment are used for subsequent self-training.

In our proposed CoM2S approach, confidence is measured by the phoneme accuracy of generated samples, and only synthetic data with a phoneme error below a predefined threshold is retained for self-training. As a result, on top of the output of the transduction model, we used a pretrained phoneme classifier [7] [13] as shown on the top left in Fig. 1. The phoneme error calculation is based on cross-entropy loss between predicted phoneme probabilities and the target phoneme sequence [10], [21]:

$$L_{\text{phoneme}} = - \sum_{t=1}^T \sum_{c=1}^C y_{t,c} \log(p_{t,c}) \quad (1)$$

where T represents sequence length (a.k.a number of phoneme steps), C number of phoneme classes (ARPABet phonemes [31]), $y_{t,c}$ ground truth one-hot encoded phoneme at step t (1 for correct class, 0 otherwise), $p_{t,c}$ the predicted probability of phoneme class c at step t . The pretrained model serves as a teacher model, guiding the selection of high-quality synthetic data generated by the EMG generator.

C. Phoneme Loss Threshold Exploration for Training Data Filtering

When working with synthetic EMG-speech pairs, data quality can vary significantly depending on how well the generated speech matches intended phonemic content. Low-quality synthetic data may introduce noise during training, hindering

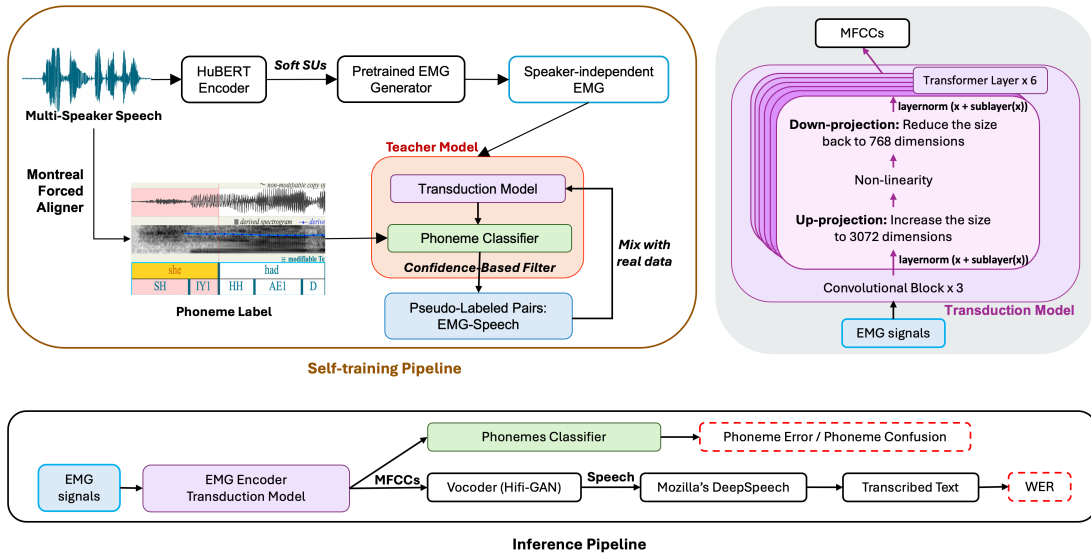


Fig. 1. Top left: Overview of our CoM2S approach for V-ETS. We employ a GAN-based EMG generator [21] conditioned on speaker-independent Soft Speech Units generated from HuBERT Encoder [24], with learnable session embeddings accounting for variations in electrode configurations. The generated EMG data then undergoes preprocessing, including upsampling and inverse transformation, to align with real EMG signals as described in Sec. II-A. A pretrained transduction model together with the pretrained classifier [10] serves as the teacher model, filtering synthetic samples based on phoneme accuracy. Only high-confidence synthetic data is retained and proportionally mixed with real EMG data for self-training, ensuring robust adaptation while maintaining phonetic consistency. Top right: baseline transduction model architecture [7], [10]. Bottom: inference pipeline.

model generalization. To address this, we explored whether filtering synthetic samples based on phoneme prediction loss could serve as an effective proxy for confidence. The core motivation was to assess whether prioritizing high-confidence examples could improve training efficiency and model performance, even at the cost of reducing the overall training data volume. This approach aims to balance the trade-off between data quantity and quality in scenarios where large-scale high-fidelity synthetic data is difficult to guarantee. Therefore, we investigate the impact of different phoneme loss thresholds on model performance.

Using the synthetic data generation approach described in Section II-A, we created training subsets by applying different phoneme loss thresholds, which were then used to train separate instances of the baseline ETS model. Performance was evaluated across various test sets to assess the relationship between training data confidence and downstream performance. Again, the goal was to determine the optimal phoneme loss threshold that balances data quality and volume, thereby improving training efficacy and downstream V-ETS performance.

D. Mix-proportion Exploration for Self-Training

To further improve training efficiency, we proportionally mix the filtered synthetic data with real EMG data, ensuring a balanced representation of both real and synthetic signals. The inclusion of real baseline EMG-speech data is crucial, as our self-training method relies on both real and synthetic sources to strike a balance between authentic V-ETS mappings and data diversity from multi-speaker synthetic samples. By integrating both, we aim to leverage the robustness of real data

while enhancing generalization with synthetic data, ensuring that the model remains grounded in real EMG patterns while benefiting from additional training examples. By carefully tuning the real-to-synthetic ratio, we aim to optimize the trade-off between model generalization and training stability. The overall pipeline of our approach is illustrated in Fig. 1. After supporting evidence, we investigate the relationship between dataset scale and self-training efficacy by progressively increasing the total training volume while maintaining the established ratio.

E. Train-from-Scratch Approach for Synthetic Data Investigation

In our previous investigation, we did not use purely real data for inference purposes because the pretrained model from [10] had already been trained on all available real voiced data. As a result, evaluating on a fully real test set would not provide meaningful insights. Additionally, to rigorously assess the contribution of synthetic EMG-speech data, we need a model where synthetic data is fully integrated into the learning process rather than used as a secondary refinement step.

To address both concerns, we propose a train-from-scratch method: using a mix of real and synthetic data based on the best mix ratio. This ensures that the voiced test dataset remains entirely unseen during training while also allowing us to directly compare the impact of synthetic data on V-ETS conversion. Unlike self-training, where synthetic data is introduced after pretraining, this approach ensures that both real and synthetic data contribute equally to the learning process from the beginning. For a fair comparison, we compared with the baseline model trained exclusively on real voiced EMG-

speech data, as described in [10]. Both models share the same architecture and training procedures to ensure consistency. By comparing WER, we determine whether synthetic data enhances model performance beyond what can be achieved with real data alone.

III. EXPERIMENTAL SETUP

A. Baseline Model and Dataset

We used the transduction model [10] as our baseline ETS model, a widely recognized baseline for ETS models. Additionally, to our knowledge, it is the only one that has trained and evaluated purely on voiced EMG, matching our setting. This model is originally designed for EMG recorded with eight electrodes, in line with the synthetic EMG data. As drawn on top right in Fig. 1, its transformer-based [32] architecture consists of three convolutional blocks followed by six transformer layers, directly processing EMG signals as well as session embedding to predict Mel-Frequency Cepstral Coefficients (MFCCs) as output.

For the real dataset, we selected the parallel and non-parallel voiced EMG-speech data from [7], excluding the silent data, as our focus is on V-ETS conversion.

B. Automatic Evaluation Metrics

To ensure a fair and controlled comparison with the baseline ETS model [8], we adopt an automatic evaluation method that isolates the impact of changes in the ETS transduction model. Specifically, both the phoneme classifier [10] and the HiFi-GAN vocoder [33] are kept identical to those used in the baseline and are frozen during all self-training or train-from-scratch experiments, ensuring that they do not adapt to any artifacts or speaker variability introduced by synthetic inputs. This design choice guarantees that any observed improvements or degradations in performance are attributable solely to the V-ETS model and not to adaptation in downstream components.

We use three automatic metrics for evaluation: phoneme accuracy, phoneme confusion, and word error rate (WER). Phoneme accuracy and confusion are computed using the frozen phoneme classifier applied to the MFCC output of the V-ETS model and are defined as follows:

$$\text{confusion}(p_1, p_2) = \frac{e_{p_1, p_2} + e_{p_2, p_1}}{f_{p_1} + f_{p_2}} \quad (2)$$

$$\text{accuracy}(p_1, p_2) = \frac{e_{p_1, p_1} + e_{p_2, p_2}}{f_{p_1} + f_{p_2}} \quad (3)$$

where e_{p_1, p_2} denotes the number of times phoneme p_2 was predicted when the ground truth was p_1 , and f_{p_1} is the total number of occurrences of phoneme p_1 in the dataset. The WER is calculated using Mozilla’s DeepSpeech [34] applied to the final speech waveform produced by the frozen HiFi-GAN vocoder. This pipeline mirrors the original evaluation setup in [8], enabling direct comparison.

By keeping the classifier and vocoder fixed and trained solely on real data, we avoid introducing evaluation bias, particularly when testing models trained with synthetic data.

This setup allows us to interpret changes in WER and phoneme metrics as genuine improvements in V-ETS transduction quality, not artifacts of downstream model adaptation.

C. Preprocessing Generated EMG for Speech Synthesis

To preprocess the generated EMG data before feeding to the transduction model, we apply an upsampling and reverse transformation procedure. First, we upsample the signal from its original sampling rate to the target rate using linear interpolation. This ensures temporal alignment with other EMG recordings at a unified frequency. Next, we apply reverse processing to restore the EMG signal to its original range. Since the GAN-generated EMG values are transformed via a tanh function during training [21], we apply an inverse tanh (arctanh) transformation to recover the original distribution. To avoid numerical singularities at extreme values (-1 and 1), we first clip the signal within the range $[-1 + 10^{-10}, 1 - 10^{-10}]$. The recovered EMG values are then scaled by a factor of 100, matching the amplitude distribution of real EMG data. This processing ensures that the generated EMG signals are comparable to the original recordings while maintaining the proper frequency characteristics.

IV. RESULTS AND DISCUSSIONS

A. Phoneme-Error-Based Filtered Synthetic Libri-EMG Data

Using the approach described in Sec. II-A, we generated three subsets of synthetic EMG-speech pairs by filtering the data using phoneme loss thresholds: no filtering (Raw), loss < 0.8 , and loss < 0.5 . Each subset was then used to continue train separate baseline models, and performance was evaluated across all test sets using WER. we then trained three versions of the baseline ETS model on synthetic training subsets using the dev-clean dataset from LibriSpeech [28] filtered at different phoneme loss thresholds:

TABLE I
FILTERED SYNTHETIC DATASET SIZE UNDER DIFFERENT CONFIDENCE THRESHOLDS

Condition	Filtered Dataset Size
Raw (no filtering)	~5.4 hours
Phoneme Loss<0.8	~5.0 hours
Phoneme Loss<0.5	~0.5 hours

The evaluation metric (lower/lighter is better) in Fig. 2 indicates that the model trained on the smallest but highest-confidence filtered subset (PL<0.5, ~0.5h) consistently achieves the best or comparable performance across all test sets, including the full raw test set (~5.4h) and the filtered subsets.

Specifically, on the raw test data, the PL<0.5 trained model attains a WER of 29.36%, outperforming the models trained on larger but less filtered datasets (42.75% for PL<0.8 and 0.48.87% for raw data). This suggests that training on high-confidence, filtered data enables the model to generalize better, despite the smaller training size. Similarly, on the filtered test sets (PL<0.8 and PL<0.5), the PL<0.5 trained model matches or slightly improves upon the performance of models trained

on larger datasets, with WER of 28.54% and 17.53% respectively, reinforcing the benefit of data quality over quantity.

In summary, these results demonstrate that filtering training data by confidence (using phoneme loss thresholds) effectively improves model generalization and performance, even when reducing training data volume significantly.

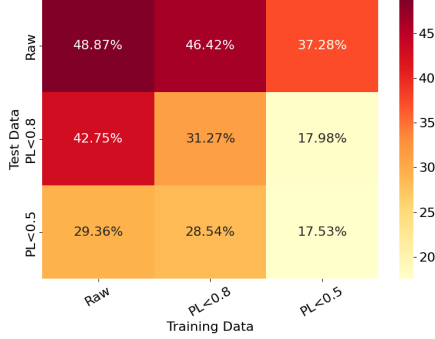


Fig. 2. Performance comparison of EMG-based speech recognition models trained on different filtered subsets of self-generated data (5.4h dev-clean in LibriSpeech [28]) and evaluated on corresponding test sets. Values and colors represent word error rates (WER) (lower/lighter is better).

TABLE II
OVERVIEW OF REAL AND SYNTHETIC VOICED DATASETS

Data	Speaker Number	Gender	Dataset Size (Utterance Number)
[8]	1	Male	7065($\approx 15.2h$)
Ours	1532	Male & Female	3514($\approx 8.3h$)

Applying the optimal phoneme-error threshold of < 0.5 , we generated 8.3 hours of EMG-speech data, covering a diverse set of 1,532 speakers across both male and female categories [35]. This multi-speaker dataset ensures robust modeling and allows us to evaluate the generator’s ability to generalize across different speakers and recording conditions. For better visualization of the dataset used in the following experiments, we list both the real baseline data discussed in Sec. III-A and the synthetic data in Table II.

B. V-ETS Performance Across Mixing Proportions and Data Quantity

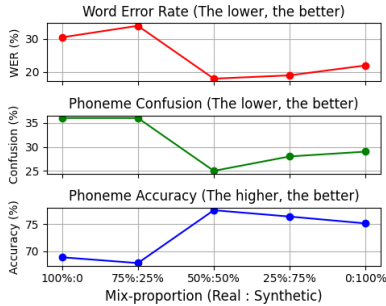


Fig. 3. The evaluation results of WER, phoneme confusion and phoneme accuracy across different real-to-synthetic data ratios.

To explore whether the effectiveness of self-training depends on the optimal mix proportion of real and synthetic data, we systematically test different training data mixing ratios to determine the optimal mix proportion of real and synthetic data. To ensure consistency, all models are trained using the same validation set drawn from real data (200 utterances). For evaluation, we construct a test set of 198 utterances, evenly split between 99 real and 99 synthetic utterances, allowing us to fairly compare the impact of different mix proportions.

The results are visualized in Figure 3, where the x-axis represents different mix proportions: 100%:0%, 75%:25%, 50%:50%, 25%:75%, and 0%:100%, corresponding to the ratio of single-speaker real EMG data to multi-speaker synthetic Libri-EMG data. For evaluation, we employ three automatic metrics: WER, phoneme confusion rate, and phoneme accuracy. The results in Fig. 3 show that the model trained with a balanced mix of 50% real and 50% synthetic data outperforms all other configurations across all three evaluation metrics.

As mentioned in Sec. II-D, we also evaluate the impact of dataset scaling on model performance under the optimal mix proportion 1:1. We trained our baseline ETS model on progressively larger datasets ($1\times$, $2\times$, and $5\times$ 1300 utterances) under the optimal 1:1 mix proportion. As shown in Table III, increasing the dataset size consistently improved the performance in terms of all metrics:

TABLE III
PERFORMANCE COMPARISON ACROSS DATASET SIZES WITH REAL AND SYNTHETIC DATA (MIX PROPORTION 1:1)

Dataset Size (50% real + 50% synthetic)	WER	Phoneme Confusion	Phoneme Accuracy
1300 utt. ($\sim 3.2h$)	23.85	29.80	74.19
$2\times$ 1300 utt. ($\sim 6.4h$)	21.88	28.57	75.34
$5\times$ 1300 utt. ($\sim 16h$)	18.03	25.45	77.59

As shown in Table III, increasing the dataset size from 3.2h to 16h of training data led to consistent improvements across all metrics: word error rate (WER) decreased by 24.4% (from 23.85% to 18.03%), phoneme confusion reduced from 29.80% to 25.45%, and phoneme accuracy improved from 74.19% to 77.59%. This indicates that increasing the dataset size enhances model robustness and generalization.

C. Scratch-Trained Model Evaluations

As discussed in Sec. II-E, we implement a controlled ablation study comparing two training paradigms: (1) a baseline model trained exclusively on real voiced EMG-speech pairs and (2) our proposed mixed-data model initialized with the previously determined optimal 1:1 real-synthetic ratio.

1) *Cross-Model Analysis with Baseline Data:* Table IV shows that our mix-train-from-scratch model achieves a WER of 21.87% on the real single-speaker test set, outperforming the previous state-of-the-art WER of 23.30% reported by the original voiced baseline [8]. This suggests that synthetic pseudo-labeled EMG data can contribute positively to model learning, even when evaluated on real, natural articulatory inputs, validating both the effectiveness of our synthetic data

TABLE IV
THE COMPARISON OF WER RESULTS ACROSS DIFFERENT MODELS AND DATASETS

Test Dataset	Baseline model [10]	Our CoM2S with self-training (mix ratio 1:1)	Voiced baseline model [8], [10]	Our CoM2S with mix-train-from-scratch (mix ratio 1:1)
Real Single-speaker Data [8], [10]	-	-	23.30% [8]	21.87%
Our multi-speaker Libri-EMG	54.21%	15.90%	37.63%	8.75%

generation and its benefits for representation learning through increased training diversity.

2) *Cross-Dataset Generalization*: As supporting evidence, we also explored model generalization by evaluating all models on these two Libri-EMG datasets. As shown in Table V, the baseline model performs poorly on this set (WER 54.21%). In contrast, our mix-train-from-scratch model achieves a WER of 8.75%, demonstrating strong generalization to speaker-independent data. Notably, even the self-training model initialized from the baseline (15.90%) surpasses the baseline model by a large margin. These results indicate that the transduction model benefits from a more diverse training set, leading to better generalization across unseen data.

D. Subjective Evaluations by Human Listeners

To complement our automatic metrics, we conducted a subjective evaluation study to assess our proposed CoM2S model in terms of speech intelligibility [10] and speech quality [36]–[38] using two scratch-trained models, with both real and synthetic data as test sets. Two representative audio samples have been made available online [39].

1) *Speech Intelligibility*: Following a similar protocol to our automated transcription tests, we engaged one human evaluator who were unfamiliar with the target utterances. The evaluator listened to 20 randomly selected synthesized speech samples and transcribed what they perceived.

TABLE V
HUMAN WER COMPARISON BETWEEN REAL AND SYNTHETIC TEST SETS

Model	WER (real)	WER (synthetic)
Voiced baseline model	27.35%	27.04%
Our CoM2S with mix-train-from-scratch	23.58%	13.57%

As shown in Table V, the mix-train-from-scratch model achieves a 15.1% relative reduction in WER on the real test set, confirming that synthetic data augmentation enhances generalization to real EMG-speech pairs. Notably, the model shows even stronger gains on synthetic test data (13.57% WER, 49.8% improvement over baseline), suggesting effective learning of synthetic patterns while maintaining real-world applicability. The persistent gap between real and synthetic performance highlights an opportunity to better align synthetic training data with real EMG characteristics in future work.

2) *Speech Quality*: MOS (Mean Opinion Score) [36], [38] is a subjective evaluation metric used to assess the perceived speech quality of synthesized or processed speech. Unlike objective metrics like WER, MOS captures human judgments of speech quality and overall listening experience. In our evaluation, ten evaluators rated the outputs on a 5-point scale (1: Bad, 5: Excellent) [36].

TABLE VI
MOS COMPARISON BETWEEN REAL AND SYNTHETIC TEST SETS

Model	MOS (real)	MOS (synthetic)
Voiced baseline model	3.00	3.45
Our CoM2S with mix-train-from-scratch	3.25	4.15

The results are shown in Table VI, the mix-train-from-scratch model achieves significantly higher MOS ratings than the real-only baseline on both real and synthetic test sets. Three key insights emerge: (1) The 8.3% improvement on real data confirms that synthetic augmentation yields perceptibly higher-quality speech despite EMG artifacts; (2) The model’s superior performance on synthetic data (20.3% higher MOS) suggests it successfully leverages multi-speaker diversity during training; (3) The baseline’s synthetic-set advantage (3.45 vs 3.00) implies inherent vocoder bias toward cleaner synthetic inputs. This improvement is notable given the frozen vocoder constraint, indicating that the gains stem primarily from the encoder’s improved EMG representation learning.

V. CONCLUSIONS

In this study, we investigate the use of synthetic EMG-speech data in self-training and enhance V-ETS model performance. Extensive experimental results confirm that our proposed CoM2S approach enhances phoneme recognition accuracy, reduces phonological confusion and word error rate, proving its effectiveness for V-ETS systems. Subjective evaluations also verify the intelligibility of the generated speech from our proposed model by human listeners. These results support the integration of synthetic data into future V-ETS training pipelines, potentially reducing reliance on large-scale real EMG recordings while maintaining high performance. Building on our previous work [40], we aim to extend the framework by introducing articulatory-level patterns derived from muscle activity in future studies.

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